Lecture 40

# Line integrals, potentials and Green

**Uses.**

Summary 4: Every bounded real sequence has a convergent subsequence; finite coordinate extraction gives this for bounded vector sequences.
Summary 8: Scalar product and chain differentiation rules hold.
Summary 12: Scalar products, norms and the triangle inequality control vectors.
Summary 26: Proper continuous integrals obey substitution and the fundamental theorem.
Summary 28: A $C^1$ parametrised curve has finite length given by the integral of its continuous speed.
Summary 36: Continuous partial derivatives imply differentiability; gradients and the matrix chain rule describe derivatives along curves.
Summary 37: A $C^2$ scalar function has equal mixed second partial derivatives.
Summary 38: M gives proper compact Fubini and parameter differentiation for an integrand jointly $C^1$ near the compact product.

**Objectives.** *Work sums the component of a force along a path. A gradient turns this sum into an endpoint difference. We prove when the cross-partial test constructs a potential, including the domain condition that the test needs. Green's theorem then relates a closed-path integral to the field throughout a planar region.*

By the end of this lecture you will be able to:

- Compute a vector-field line integral with its path orientation and prove the gradient endpoint formula.
- Characterise path independence by a potential on an open polygonally connected domain.
- Construct a potential for a $C^1$ field with symmetric cross partials on an open star-shaped domain, and explain the punctured-plane guard.
- Prove and apply Green's theorem for the stated compact graph regions, with positive boundary orientation.

## 40.1 Work along an oriented path

Let $U \subset \mathbb{R}^n$ be open and let $\boldsymbol{F} : U \to \mathbb{R}^n$ be continuous. An oriented piecewise $C^1$ path is a continuous map $\boldsymbol{\gamma} : [a, b] \to U$ with finitely many $C^1$ pieces; increasing parameter gives its

direction. Its vector-field line integral is

$$\int_\gamma \boldsymbol{F}\cdot \mathrm{d}\boldsymbol{x} = \int_a^b \boldsymbol{F}(\boldsymbol{\gamma}(t))\cdot \boldsymbol{\gamma}'(t)\,\mathrm{d}t,$$

with a sum over those pieces if needed. For a force field this is work: a displacement contributes only the component of force along that displacement. The notation $\mathrm{d}\boldsymbol{x}$ packages the coordinate differentials; it is not an extra scalar length factor. In the plane, for $\boldsymbol{F}=(P,Q)$, the same integral is written $\int_\gamma P\,\mathrm{d}x + Q\,\mathrm{d}y$.

The chain rule and one-variable substitution prove invariance under an increasing $C^1$ change of parameter whose inverse is $C^1$ on each piece. Indeed, writing $t=h(s)$ gives velocity $\boldsymbol{\gamma}'(h(s))h'(s)$, exactly the substitution factor in the scalar integral. Reversing the parameter changes the sign. Joining two paths adds their integrals, by splitting the parameter integral at the joining point. Thus orientation matters, whereas the speed of traversal does not under these admissible changes of parameter. A closed path has the same initial and final point; its integral is called *circulation* and is often written with $\oint$.

**Example 40.1** (A nonuniform force along a curved path). Take

$$\boldsymbol{F}(x,y,z)=(2xy,x^2,e^z),\qquad \boldsymbol{\gamma}(t)=(t,t^2,\ln(1+t)),\qquad 0\le t\le 1.$$

The field is continuous on $\mathbb{R}^3$ and the path is $C^1$. Direct substitution gives

$$\boldsymbol{F}(\boldsymbol{\gamma}(t))=(2t^3,t^2,1+t),\qquad \boldsymbol{\gamma}'(t)=(1,2t,1/(1+t)).$$

Their scalar product is $4t^3+1$, hence the work is

$$\int_\gamma \boldsymbol{F}\cdot\mathrm{d}\boldsymbol{x}=\int_0^1 (4t^3+1)\,\mathrm{d}t = 2.$$

The velocity factor is essential: integrating the field components alone would not measure displacement along this curve.

## 40.2 Gradients and endpoint differences

A *potential* for $\boldsymbol{F}$ is a $C^1$ scalar function $\phi$ with $\boldsymbol{F}=\nabla\phi$ on the domain under discussion. We use the mathematical sign convention $\boldsymbol{F}=\nabla\phi$. A mechanical potential energy $V$ instead gives $\boldsymbol{F}=-\nabla V$.

**Theorem 40.2** (Gradient endpoint formula). *If $\phi\in C^1(U)$ and $\gamma$ is a piecewise $C^1$ path in $U$ from $A$ to $B$, then*

$$\int_\gamma \nabla\phi\cdot\mathrm{d}\boldsymbol{x}=\phi(B)-\phi(A).$$

*Proof.* On each $C^1$ piece, the chain rule gives $(\phi\circ\boldsymbol{\gamma})'(t)=\nabla\phi(\boldsymbol{\gamma}(t))\cdot\boldsymbol{\gamma}'(t)$. The fundamental theorem identifies its integral with the endpoint difference on that piece. Adding the finitely many differences cancels all internal endpoint values and leaves the claimed expression. ∎

For the preceding example, $\phi=x^2y+e^z$ is a potential. Its endpoint values are 3 and 1, agreeing with the direct work 2. For a mechanical potential, apply the theorem to $-V$ to obtain

$$\int_\gamma \boldsymbol{F}\cdot\mathrm{d}\boldsymbol{x}=V(A)-V(B).$$

For instance, a spring force $F(x) = -kx$, $k > 0$, has energy $V(x) = kx^2/2$; its work from 0 to $b$ is $-kb^2/2$. A gradient has zero circulation along every closed path in its domain.

We say work is *path independent* if all piecewise $C^1$ paths with the same endpoints have the same integral. On a domain where such paths join the points in question, this is equivalent to zero circulation along every closed path. To prove this, join one path to the reverse of the other: its closed integral is their difference. Conversely, a closed path has coincident endpoints, and comparison with the constant path gives zero.

There is also a converse construction of a potential. Assume $U$ is open and *polygonally connected*, meaning that any two points can be joined by a finite broken line lying in $U$. Let $\boldsymbol{F}$ be continuous and its work path independent. Fix $A \in U$ and define $\phi(x)$ to be the work along any such path from $A$ to $x$. This is well-defined by path independence. For $x \in U$, take a ball about $x$ contained in $U$. For sufficiently small positive or negative $h$, join the selected path to the coordinate segment from $x$ to $x + he_j$. Additivity and the straight-segment parametrisation give

$$\phi(x + he_j) - \phi(x) = \int_0^h F_j(x + te_j)\,\mathrm{d}t.$$

Dividing by $h$ and using continuity gives $\partial_j \phi(x) = F_j(x)$. These partials are continuous, so Lecture 36 proves $\phi \in C^1(U)$ and $\nabla\phi = \boldsymbol{F}$. Thus, under the stated domain hypothesis, a continuous field is path independent exactly when it has a potential. No theorem identifying different notions of connectedness was used.

On such a domain, two potentials for the same field differ by one constant. Indeed, their difference has zero gradient. Join any point to a fixed base point by a broken line in the domain and apply the gradient endpoint formula to this difference. Its integral is zero, so its two endpoint values agree. This also explains why an arbitrary additive constant in a computed potential has no effect on work.

## 40.3 The cross-partial test on a star-shaped domain

A set $U$ is *star-shaped about* $a \in U$ if the whole segment $a + t(x - a)$, $0 \le t \le 1$, lies in $U$ for every $x \in U$. Figure 40.1 contrasts the star-shaped segment condition with a closed path around an excluded point.

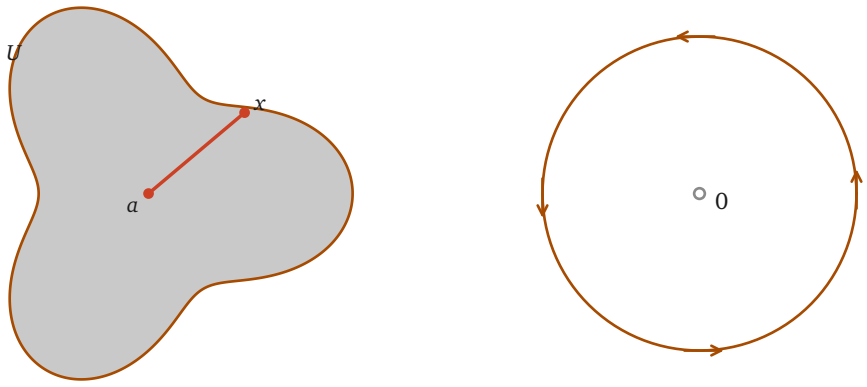



Figure 40.1: A star-shaped region contains every segment from its centre $a$ to a point $x$. The second panel is an anticlockwise circle around an omitted origin, the geometry of the later domain example.

Such a set is polygonally connected: travel from one point to $a$, then to the other. A ball is star-shaped about its centre by the norm inequality.

For a $C^1$ field to have a potential, symmetric cross partials are necessary. Indeed, $\partial_j \phi = F_j \in C^1$ implies $\phi \in C^2$, so Schwarz gives $\partial_i F_j = \partial_j F_i$. In dimension three these three independent equalities are written $\operatorname{curl} \boldsymbol{F} = \boldsymbol{0}$, where, componentwise,

$$\operatorname{curl} \boldsymbol{F} = (\partial_y F_3 - \partial_z F_2,\ \partial_z F_1 - \partial_x F_3,\ \partial_x F_2 - \partial_y F_1).$$

In the plane, the scalar curl is $\partial_x Q - \partial_y P$. We prove sufficiency with a precise domain assumption.

**Theorem 40.3** (Potential on a star-shaped domain). *Let $U \subset \mathbb{R}^n$ be open and star-shaped about $a$. Let $\boldsymbol{F} \in C^1(U)$ satisfy $\partial_i F_j = \partial_j F_i$ for all $1 \le i, j \le n$. Then a potential is*

$$\phi(x) = \int_0^1 \boldsymbol{F}(a + t(x - a)) \cdot (x - a)\, \mathrm{d}t.$$

*Proof.* First justify differentiating this proper integral. Fix $x \in U$ and choose a ball $B$ about $x$ whose closure lies in $U$. Star-shapedness puts $a + t(z - a)$ in $U$ for every $(z, t)$ in $K = \overline{B} \times [0, 1]$. We need this property on an open neighbourhood of $K$, including small extensions of the integration endpoints. If no such neighbourhood existed, for every integer $k \ge 1$ we could choose $(w_k, s_k) \in K$ and $(z_k, t_k)$ at distance less than $1/k$ from it, with $a + t_k(z_k - a) \notin U$. The bounded coordinates of $(w_k, s_k)$ have a convergent subsequence, by successive applications of Lecture 4. Its limit $(w_*, s_*)$ lies in $K$, since the ball closure and the interval are closed. The nearby sequence $(z_k, t_k)$ has the same limit along that subsequence. Continuity of the affine path map gives $a + t_k(z_k - a) \to a + s_*(w_* - a) \in U$. Openness of $U$ forces these points eventually into $U$, a contradiction. The integrand is therefore jointly $C^1$ on a neighbourhood

of the whole compact product: it composes a $C^1$ field with this affine map and a scalar product.

The parameter clause of standing fact M says that a proper integral on a fixed compact interval may be differentiated when its integrand is jointly $C^1$ on a neighbourhood of this compact parameter–interval product. We have just checked this clause, so for $h = x - a$ it gives

$$\partial_j \phi(x) = \int_0^1 \left( F_j(a + th) + t \sum_{i=1}^{n} h_i \partial_j F_i(a + th) \right) \mathrm{d}t.$$

Replace $\partial_j F_i$ by $\partial_i F_j$ using the hypothesis. The resulting integrand is exactly

$$\frac{\mathrm{d}}{\mathrm{d}t}\big(tF_j(a + th)\big) = F_j(a + th) + t \sum_{i=1}^{n} h_i \partial_i F_j(a + th).$$

The fundamental theorem gives $\partial_j \phi(x) = F_j(x)$, since the endpoint at zero has factor $t = 0$. The resulting partials are continuous, so $\phi \in C^1(U)$ and is the required potential. This proves the theorem from M [P]. ∎

The radial formula establishes existence. When computing a particular potential, integrating one component and then determining the missing functions is often shorter. For example, consider on $\mathbb{R}^3$ the family

$$\boldsymbol{F} = (2xy + z,\ x^2 + \alpha z,\ \beta x + y), \qquad \alpha, \beta \in \mathbb{R}.$$

The first cross-partial pair already agrees. The other two require $1 = \beta$ and $\alpha = 1$. If either fails, necessity rules out a potential on any nonempty open subdomain. If both hold, the star-shaped theorem guarantees a potential. To find one, integrate $\partial_x \phi = 2xy + z$:

$$\phi = x^2 y + xz + C(y, z).$$

The $y$ component requires $\partial_y C = z$, hence $C = yz + D(z)$; the $z$ component then requires $D' = 0$. Thus $\phi = x^2 y + xz + yz$ works, as direct differentiation verifies. Its work from $(0, 0, 0)$ to $(1, 2, -1)$ is $-1$, along every admitted path.

## 40.4 Why the domain cannot be omitted

On $U = \mathbb{R}^2 \setminus \{(0, 0)\}$ consider

$$\boldsymbol{F}(x, y) = \left( -\frac{y}{x^2 + y^2}, \frac{x}{x^2 + y^2} \right).$$

It is $C^1$ on this open domain, and differentiation gives

$$\partial_y F_1 = \frac{y^2 - x^2}{(x^2 + y^2)^2} = \partial_x F_2.$$

Nevertheless, on the positively traversed circle $\gamma(t) = (b \cos t, b \sin t)$, $0 \le t \le 2\pi$, $b > 0$,

$$\boldsymbol{F}(\gamma(t)) = \frac{1}{b}(-\sin t, \cos t), \qquad \boldsymbol{\gamma}'(t) = b(-\sin t, \cos t), \qquad \oint_\gamma \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{x} = 2\pi.$$

A global potential would give zero by the endpoint formula, so none exists. The punctured plane is not star-shaped about any of its points: for a chosen $a \neq 0$, the segment to $-a$ meets the omitted origin. Each point does have a small ball avoiding the origin, and on that ball the theorem supplies a local potential. Local potentials do not imply a single potential on the whole punctured domain.

## 40.5 Green's theorem for a graph region

For a planar region, positive boundary orientation keeps the region on the left as the boundary is traversed. An outer circular boundary is consequently anticlockwise. Assume $D$ is a compact region simple in both coordinate directions:

$$D = \{a \le x \le b,\ g_1(x) \le y \le g_2(x)\},$$
$$D = \{c \le y \le d,\ h_1(y) \le x \le h_2(y)\}.$$

Figure 40.2 shows the positive orientation and a vertical slice.

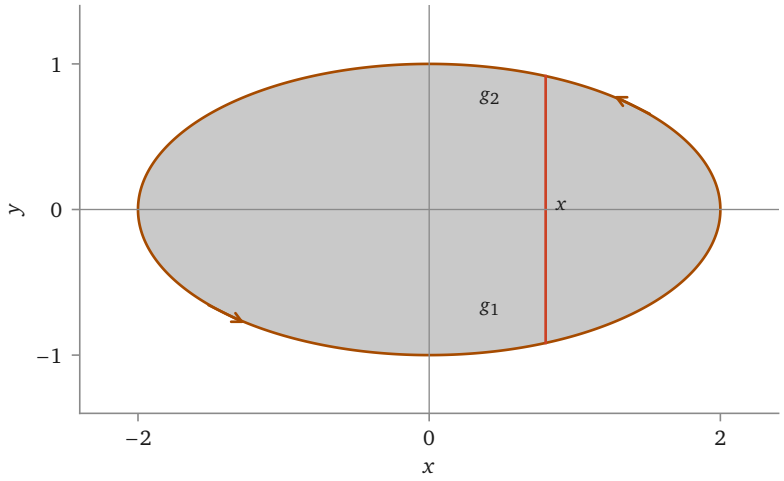



Figure 40.2: Positive traversal goes rightwards along the lower graph and leftwards along the upper graph. A vertical slice joins $g_1(x)$ to $g_2(x)$.

The graph bounds are continuous and piecewise $C^1$ away from possible endpoint vertical or horizontal tangencies. The boundary itself is a simple closed piecewise $C^1$ curve, and $D$ satisfies M. Here simple closed means that the boundary can be traced once without self-intersections, with only the initial and final points coinciding. For these graph descriptions, positive traversal runs rightwards along the lower graph and leftwards along the upper graph; equivalently, it runs upwards along the right graph and downwards along the left graph. These hypotheses include discs and ellipses; their graph derivatives may be unbounded at the outermost coordinate values, while their boundary parametrisations remain smooth.

**Theorem 40.4** (Green's theorem on a region simple in both directions)**.** *Let $D$ have the preceding properties, let $\partial D$ have positive orientation, and let $P, Q$ be $C^1$ on an open neighbourhood of $D$. Then*

$$\oint_{\partial D} P\,\mathrm{d}x + Q\,\mathrm{d}y = \iint_D (\partial_x Q - \partial_y P)\,\mathrm{d}A.$$

*Proof.* The lower graph is traversed from left to right and the upper graph from right to left. Vertical end pieces have $\mathrm{d}x = 0$. The definition of the line integral therefore gives

$$\oint_{\partial D} P\,\mathrm{d}x = \int_a^b \big(P(x, g_1(x)) - P(x, g_2(x))\big)\mathrm{d}x.$$

At an endpoint tangency, first use closed interior subintervals of the graph parameters and then take the endpoint limit. The missing boundary arc has length tending to zero by its

piecewise $C^1$ parametrisation. Since $P$ is bounded on that arc, its contribution tends to zero. The right-hand integrand is continuous up to the endpoints, giving the same displayed proper integral. The fundamental theorem on each vertical segment and compact Fubini yield

$$\oint_{\partial D} P\,\mathrm{d}x = -\int_a^b \int_{g_1(x)}^{g_2(x)} \partial_y P(x,y)\,\mathrm{d}y\,\mathrm{d}x = -\iint_D \partial_y P\,\mathrm{d}A.$$

On the right graph positive orientation is upwards, and on the left it is downwards. Horizontal end pieces have $\mathrm{d}y = 0$. Repeating the same argument with horizontal slices gives

$$\oint_{\partial D} Q\,\mathrm{d}y = \int_c^d \big(Q(h_2(y),y) - Q(h_1(y),y)\big)\mathrm{d}y = \iint_D \partial_x Q\,\mathrm{d}A.$$

Add the two identities. All double integrals are proper, and the derivatives are continuous on a neighbourhood of the entire compact region [P from M]. ∎

**Example 40.5** (Circulation around an ellipse)**.** Let $D = \{x^2/4 + y^2 \le 1\}$ and $\boldsymbol{F} = (-y^3, x^3)$. The polynomial field is $C^1$ on the whole plane, and the ellipse satisfies the theorem. Its positive boundary circulation is

$$\oint_{\partial D} -y^3\,\mathrm{d}x + x^3\,\mathrm{d}y = 3\iint_D (x^2+y^2)\,\mathrm{d}A.$$

Set $x = 2u$, $y = v$ on the unit disc, with absolute Jacobian 2. Polar integration gives $\iint_D x^2\,\mathrm{d}A = 2\pi$ and $\iint_D y^2\,\mathrm{d}A = \pi/2$: the radial factor is $\int_0^1 r^3\,\mathrm{d}r = 1/4$, and each squared angular factor integrates to $\pi$. Thus the circulation is $15\pi/2$. A direct orientation check uses $x = 2\cos t$, $y = \sin t$ for one increasing turn: the integrand becomes $2\sin^4 t + 8\cos^4 t$. Each fourth-power integral over a full turn is $3\pi/4$, from the half-angle identities, again giving $15\pi/2$.

For finite unions of suitable pieces, the extension is a sketch [S]: apply the proved identity on the pieces and cancel the oppositely traversed internal boundaries. A region with holes has positive outer boundaries and clockwise inner boundaries, so the remaining orientations keep the region on the left. We do not label the general geometric subdivision argument as a full proof. For the punctured-plane field, Green cannot be applied to a whole disc, because the field is undefined at its centre. On an annulus, the extension [S] has zero scalar curl and total circulation $2\pi - 2\pi = 0$, with the inner boundary traversed clockwise. This agrees with the directly computed circle integrals and does not remove their nonzero individual values.

**Summary 40.**

- A continuous field's oriented work integral is the integral of its scalar product with path velocity. Reversing orientation reverses its sign.
- A $C^1$ potential gives work $\phi(B) - \phi(A)$; mechanical energy with $\boldsymbol{F} = -\nabla V$ gives $V(A) - V(B)$. On an open polygonally connected domain, a continuous field is path independent exactly when it has a potential.
- Symmetric cross partials suffice for a $C^1$ field on an open star-shaped domain, by Theorem 40.3. The punctured-plane circulation shows why a domain hypothesis is necessary.
- For the stated compact regions simple in both directions and a field $C^1$ near the whole region, positive boundary circulation is the area integral of scalar curl. Finite unions and holes use the stated sketch extension [S].

Sheet 40 develops work, local and global potentials, exact differential equations and further Green calculations. Lecture 41 interprets planar flux and then defines flux across oriented surfaces.

## References

J. Stewart, D. Clegg and S. Watson, *Calculus: Early Transcendentals*, 9th ed., Cengage, 2020, §§16.1–16.4.